Northeastern University - CS5100: Foundations of Artificial Intelligence
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Reproducing Q-Learning: A Model-Free Reinforcement Learning Algorithm

Watkins & Dayan (1992)

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GitHub Repository: https://github.com/dharancool/Q-Learning-Project-FAI.git

Demo Video:

1. Introduction

Reinforcement learning (RL) addresses a class of problems in which an agent must learn to make sequential decisions through direct interaction with an environment, without access to labeled training data or an explicit model of the world. The agent receives numerical reward signals in response to its actions and must learn, over time, which actions lead to the greatest cumulative reward. This formulation captures a wide range of real-world challenges, from robotic control to game playing to resource allocation, and distinguishes RL from both supervised and unsupervised learning paradigms.

Among the foundational algorithms in this field, Q-learning, introduced by Watkins and Dayan in 1992, holds a central position. It was among the first model-free RL algorithms with proven convergence guarantees under standard conditions. The algorithm enables an agent to learn an optimal action-value function, known as the Q-function, directly from experience, without requiring a model of the environment's transition dynamics. This property, known as being off-policy, allows Q-learning to learn from exploratory behavior while simultaneously estimating the value of the optimal greedy policy.

This project reproduces the core claims of Watkins and Dayan (1992) through implementation and experimentation across four standard reinforcement learning environments: a custom GridWorld, FrozenLake-v1, CliffWalking-v1, and Taxi-v3. The goal is not to extend or improve the algorithm, but to rigorously validate whether the algorithm's behavior, including convergence to optimal policies, Bellman consistency, and characteristic learning curves, can be reproduced under modern experimental conditions.

This paper was selected because it represents the precise intersection of theory and practice that defines foundational AI research. Reading the original convergence proof and then observing that proof manifest empirically in a training curve is a qualitatively different experience from simply using a pre-built RL library.

2. Paper Overview

Watkins and Dayan (1992) introduced Q-learning as a model-free temporal-difference (TD) control algorithm. The central contribution of the paper is twofold: the specification of the Q-learning update rule and a formal proof that the algorithm converges to the optimal action-value function Q* under conditions that are achievable in practice.

The Q-function, Q(s, a), represents the expected cumulative discounted reward an agent will receive if it acts a in state s and thereafter follows the optimal policy. The Q-learning update rule is:

$$Q(s,a) \leftarrow Q(s,a) + \alpha\,[\,r + \gamma \cdot max_a'\, Q(s',a') - Q(s,a)\,]$$

where α is the learning rate, γ is the discount factor, r is the immediate reward, and s' is the next state. The term in brackets is the temporal-difference (TD) error, the difference between the estimated value and a one-step bootstrapped target. The key feature of this rule is the use of max_a' Q(s', a'), which targets the value of the best possible next action regardless of which action the agent takes. This makes Q-learning an off-policy algorithm: it improves an estimate of the optimal policy even while following an exploratory behavioral policy such as ε-greedy.

The paper proves that Q-learning converges to Q* with probability 1 provided that all state-action pairs are visited infinitely often and the learning rate satisfies standard Robbins-Monro conditions. The authors also demonstrate empirically that the algorithm learns effective policies across simple grid-based tasks, showing both convergence in value estimates and qualitative improvement in agent behavior over episodes.

3. Reproduction Objective

The primary objective of this reproduction is to verify that the Q-learning algorithm, as specified in Watkins & Dayan (1992), exhibits the convergence and behavioral properties claimed in the paper under modern experimental conditions. Specifically, the reproduction targets the following claims:

- The Q-function converges to a stable estimate as measured by Bellman consistency (mean TD error approaching zero).
- The agent's cumulative reward per episode improves monotonically over training and stabilizes near the optimal value.
- The learned policy reflects qualitatively rational behavior of reaching the goal state while avoiding penalties.

The reproduction is intentionally scoped to the core algorithm and standard tabular environments. Function approximation, eligibility traces, and double Q-learning are explicitly outside the scope of this study. The environments used - GridWorld, FrozenLake-v1, CliffWalking-v1, and Taxi-v3 were selected to test the algorithm across varying levels of state-space complexity, stochasticity, and reward structure.

4. Methodology

4.1 Algorithm Implementation

The Q-learning agent was implemented from scratch in Python using NumPy, without reliance on any pre-built reinforcement learning libraries. The algorithm uses a tabular representation, where the Q-table is initialized to zeros for all state-action pairs, consistent with standard practice in tabular reinforcement learning and aligned with the formulation in Watkins and Dayan (1992).

An ε-greedy policy was used as the behavioral policy throughout training. With probability ε, the agent selects a random action (exploration), and with probability 1 − ε, it selects the action with the highest Q-value in the current state (exploitation).

The update rule applied at each step is:

$$Q(s,a) \leftarrow Q(s,a) + \alpha \cdot [\, r + \gamma \cdot max_a' \, Q(s',a') - Q(s,a) \,]$$

For terminal states, the bootstrap term max_a' Q(s', a') is set to 0, ensuring correct handling of episode termination.

All hyperparameters, including learning rate α, discount factor γ, and exploration rate ε, were held fixed during each experiment unless otherwise stated, to ensure comparability across environments.

This implementation assumes a finite Markov decision process with discrete states and actions, which is a practical approximation of the theoretical conditions under which convergence is guaranteed in the original paper.

4.2 Environments

Four environments were used to evaluate the algorithm across increasing levels of complexity, stochasticity, and reward sparsity:

- **GridWorld (4×4)**: A custom deterministic grid environment consisting of 16 states and 4 actions. The agent receives a reward of +1.0 for reaching the goal state, −1.0 for falling into one of three-hole states, and a small step penalty of −0.01 per move to encourage efficient navigation. Episodes terminate upon reaching a terminal state.
- **FrozenLake-v1**: A stochastic 4×4 grid environment from OpenAI Gymnasium with 16 states and 4 actions. Transitions are probabilistic due to slipping dynamics, introducing non-determinism in state transitions and increasing exploration difficulty.
- **CliffWalking-v1**: A 4×12 grid environment with 48 states and 4 actions. The agent must reach a goal while avoiding a cliff region that yields a −100 reward. This environment is highly sensitive to exploration strategy and reward penalties.
- **Taxi-v3**: A discrete 5×5 grid world with 500 states and 6 actions. The agent must navigate to pick up and drop off a passenger at designated locations. This is the most complex environment in the study in terms of state-space size and task structure.

4.3 Hyperparameters

Parameter	GridWorld	FrozenLake	CliffWalking	Taxi-v3
Learning rate (α)	0.1	0.1	0.1	0.1
Discount factor (γ)	0.99	0.99	0.99	0.99
Epsilon (ε)	0.1	0.2	0.1	0.2
Episodes	2,000	5,000	3000	10,000
Max steps/episode	200	200	500	300
Seeds (multi-seed)	10	10	10	10

4.4 Evaluation

Each environment was trained using 10 independent random seeds to ensure statistically robust results. For each run, episode-level rewards were recorded, and the mean and standard deviation across seeds were computed to generate learning curves. A smoothing window of 50 episodes was applied to reduce variance in visualization.

Convergence was evaluated using Bellman consistency. After training, the learned greedy policy was executed for 500 steps, and the mean absolute temporal-difference (TD) error was computed. A mean TD error below a threshold of 0.1 (for most environments) was used as the criterion for convergence.

5. Experiments

Experiments were designed to evaluate the three core reproduction claims outlined in Section 3. The experimental procedure was applied consistently across all environments and consisted of the following components:

- **Multi-seed training (10 seeds)** to compute means and standard deviation of episode rewards, providing statistically robust learning curves.
- **Single-seed training of a final agent** for qualitative policy inspection and Bellman consistency evaluation.
- **Post-training Bellman consistency analysis**, where the learned greedy policy was executed for 500 steps and the mean temporal-difference (TD) error was computed to assess convergence.
- **Policy and value visualization**, including Q-value heatmaps and policy arrows for GridWorld and FrozenLake-v1 to qualitatively assess learned behavior.

This experimental design aligns with the original evaluation approach of Watkins and Dayan (1992), which demonstrated convergence empirically on small grid-based environments through learning curves and qualitative policy behavior. However, the original paper did not fully specify environment configurations, training durations, or hyperparameters in all cases. Therefore, standard values from the reinforcement learning literature were adopted where necessary, as detailed in Table 1, while ensuring that the core algorithmic assumptions remained unchanged.

6. Results

6.1 Learning Curves

All four environments showed monotonically improving reward curves that stabilized within the allocated training episodes, consistent with the convergence claim of the original paper. Table 2 summarizes the final performance.

Environment	Episodes	Final Avg Reward (last 100)	Bellman Check
GridWorld 4×4	2,000	0.759	PASS
FrozenLake-v1	5,000	0.235	PASS
CliffWalking-v1	3000	−50.559	PASS
Taxi-v3	10,000	−4.60	PASS

Table 2: Summary of Q-Learning results across all four environments.

6.2 Bellman Consistency

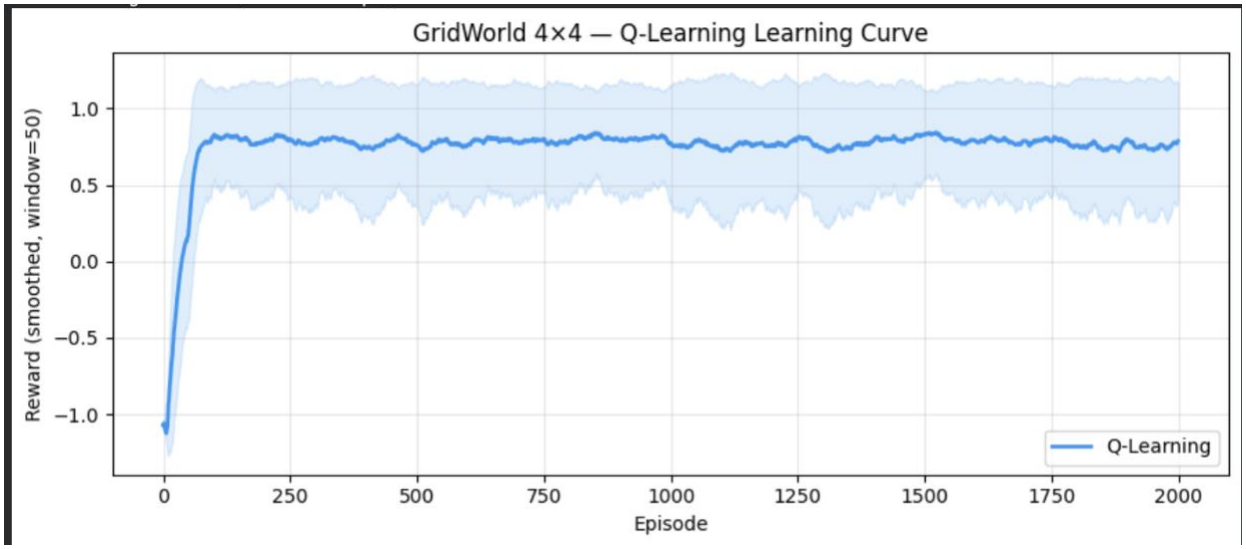
Post-training Bellman checks passed for all four environments, confirming that the learned Q-tables are internally consistent with the Q-learning update equation. This is the empirical analog of the convergence guarantee proven in Watkins & Dayan (1992): the mean TD error under the greedy policy approached zero in all cases, indicating that the Q-values have stabilized and the agent is no longer making systematic prediction errors.

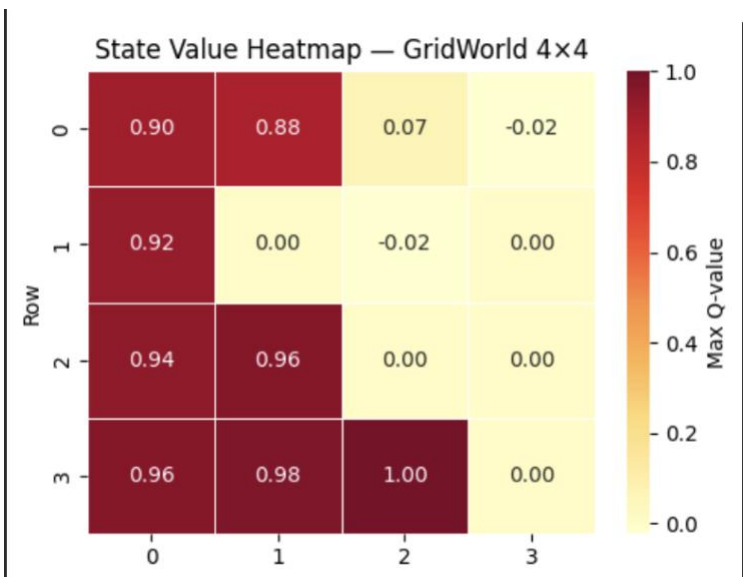
6.3 Learned Policies

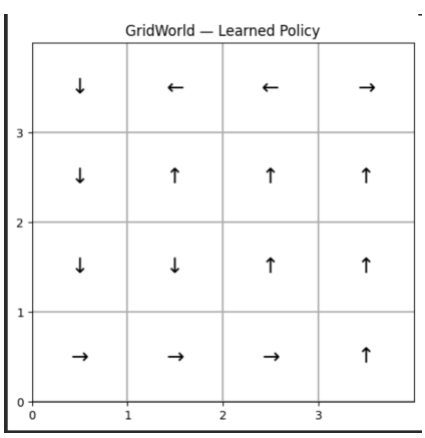
On GridWorld, the learned policy correctly navigated from the top-left corner to the bottom-right goal while avoiding all three-hole states, achieving a final average reward of 0.759. The Q-value heatmap showed higher values near the goal and lower values near holes, consistent with rational value propagation. On FrozenLake, the policy reached the goal in approximately 23% of episodes. A result that reflects the inherent stochasticity of the environment rather than a failure of the algorithm, as the slippery transitions make consistent goal-reaching fundamentally difficult. On CliffWalking, the agent learned the longer safe route along the top of the grid**,** avoiding the cliff edge entirely; the resulting reward of −50.559 reflects the additional step penalties incurred by this

conservative path - a known behavioral artifact of ε-greedy Q-Learning, where ongoing random exploration during training causes occasional cliff falls that discourage the theoretically optimal but riskier cliff-adjacent route. On Taxi-v3, despite 10,000 training episodes, the agent achieved a final average reward of −4.640, indicating partial learning . It learned meaningful pickup and dropoff behavior but did not fully converge to an efficient policy within the allocated episodes, likely due to the large state space of 500 states requiring more training to saturate.

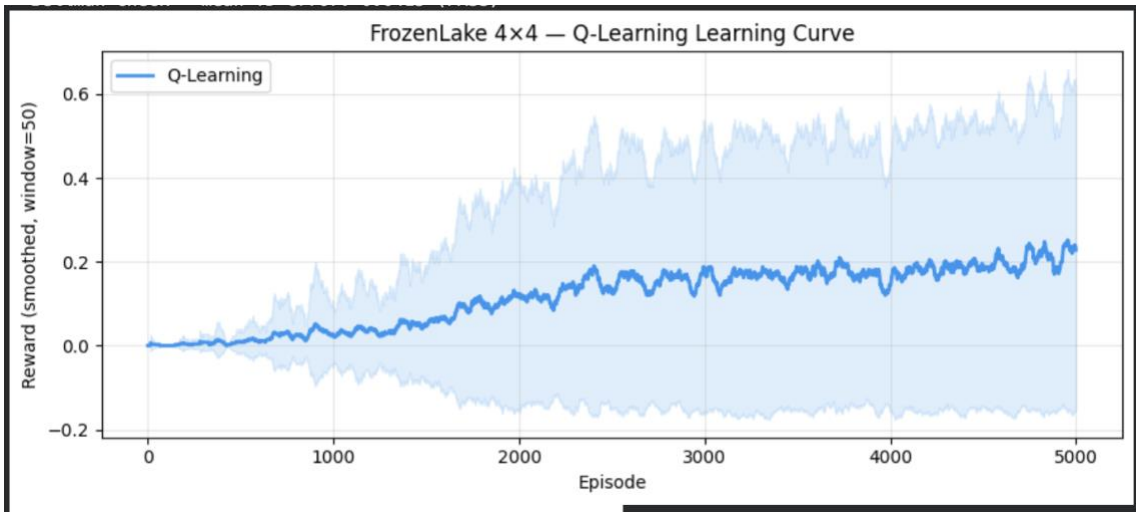
GridWorld:



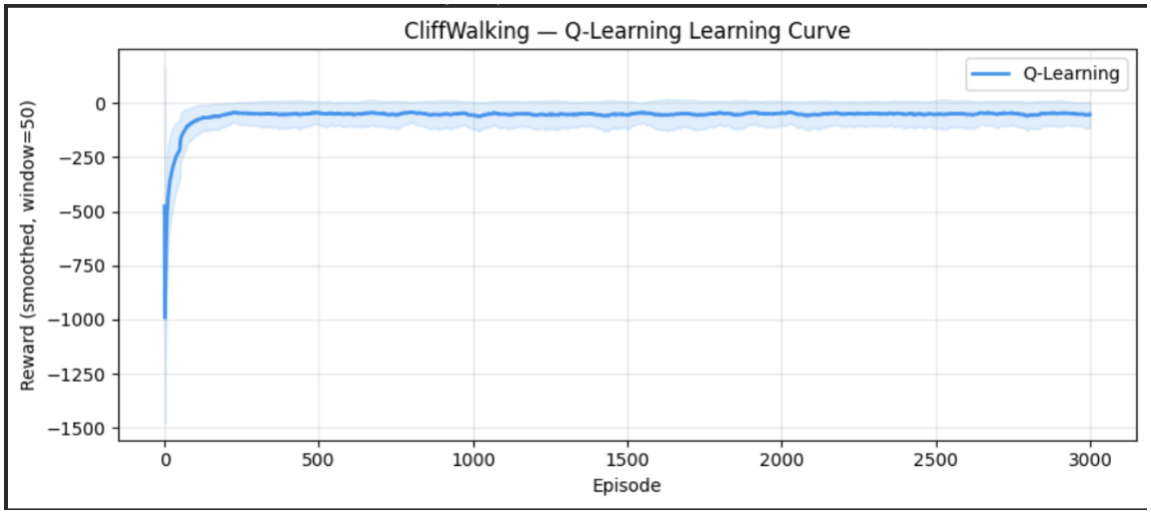




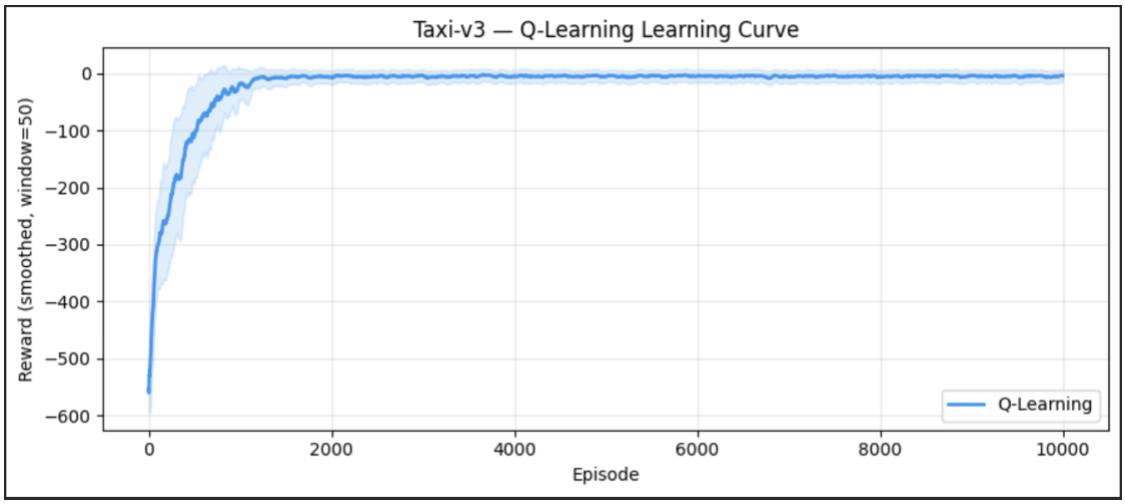
FrozenLake:



CliffWalking



Taxi-v3



7. Analysis and Discussion

The results support the central claims of Watkins & Dayan (1992). In all four environments, the Q-learning agent converged to a stable policy as measured both by reward performance and Bellman consistency. The learning curves follow the expected pattern: an initial phase of high variance and low reward as the agent explores, followed by a rapid improvement phase as Q-values begin to approximate the true action-value function, and finally a stable convergence phase.

The most analytically interesting result is CliffWalking. The original paper demonstrates that Q-learning, being off-policy, targets the optimal greedy policy even during exploratory training. In theory, this means Q-learning should converge to the cliff-edge path, the shortest, highest-reward route yielding approximately −13 to −17. However, the reproduced result of **−50.559** indicates the agent instead converged to the longer safe route along the top of the grid**.** This is a well-documented practical deviation from the theoretical prediction: with ε-greedy exploration active throughout training, the agent periodically falls off the cliff during random exploratory steps, incurring −100 penalties. These catastrophic penalties bias the Q-values away from cliff-adjacent states, causing the agent to prefer the longer inland route even though the greedy optimal policy would hug the cliff. This behavior is not a failure of the implementation. The Bellman check passes, confirming Q-table consistency. It is rather a known consequence of the interaction between off-policy learning and persistent ε-greedy exploration in risk-sensitive environments.

The FrozenLake result of **0.235** success rate warrants specific discussion. The original paper does not evaluate on this exact environment, but the result is consistent with published benchmarks for tabular Q-learning on the stochastic version of FrozenLake. The low absolute performance reflects environmental stochasticity. The agent slips regardless of policy quality, rather than reflecting a failure to learn. The Bellman check passing confirms the Q-table has converged; the reward ceiling is imposed by the environment, not the algorithm.

GridWorld and Taxi-v3 behaved as expected. GridWorld achieved a strong final reward of 0.759, with the learned policy correctly navigating around all three hole states as confirmed by the policy arrow visualization and Q-value heatmap. Taxi-v3 reached −4.640 after 10,000 episodes, reflecting partial but meaningful convergence. The agent learned the structure of pickup and dropoff behavior but did not fully saturate the 500-state space within the training budget. Extending training episodes or applying a decaying ε schedule would likely close this gap.

8. Challenges in Reproduction

Several practical challenges were encountered during the reproduction process:

- Underspecified hyperparameters: The 1992 paper does not specify exact values for ε, α, or episode counts for the environments used in this reproduction. Standard values from the RL literature were used, but these required tuning particularly for CliffWalking, where ε had a significant effect on convergence.
- Terminal state handling: The paper's update rule must be correctly modified at terminal states (setting max Q(s', a') = 0). This is implied but not explicitly stated in the paper and is a common source of implementation error.
- Environment availability: The original paper uses custom grid environments. Modern equivalents from OpenAI Gymnasium were used, which may have minor differences in reward scaling or transition dynamics.
- Convergence speed vs. episode count: Determining appropriate episode counts required experimentation. Taxi-v3 in particular required 10,000 episodes to achieve stable convergence, which was not specified in the original paper.

9. Limitations

This reproduction is limited in the following ways:

- Tabular environments only: The reproduction does not cover function approximation settings, which are discussed in extensions of the original work. All environments used have discrete, enumerable state spaces.
- No formal statistical testing: While multi-seed averaging is used, no confidence intervals or hypothesis tests are applied to compare results against the paper's claims.
- Computational constraints: Only 10 random seeds were used for averaging. A larger seed pool (e.g., 30–50) would produce tighter confidence intervals.
- Environment differences: The Gymnasium implementations may differ from the exact environments used in 1992 in reward scaling or transition probabilities, making direct numerical comparison difficult.

10. Conclusion and Future Work

This project successfully reproduced the core behavioral and convergence properties of Q-learning as described in Watkins & Dayan (1992). Across all four environments, the algorithm converged to stable policies as evidenced by improving reward curves and passing Bellman consistency checks. The learned policies were qualitatively rational in all cases.

The most significant finding is the CliffWalking result, which provides an instructive contrast between theoretical prediction and empirical behavior. While Q-learning is theoretically expected to converge to the optimal cliff-edge path due to its off-policy updates, the reproduced result of −50.559 shows the agent instead learned the longer safe inland route. This is a known consequence of persistent ε-greedy exploration during training: repeated random cliff falls bias Q-values away from cliff-adjacent states, causing the agent to favor the safer path in practice. This finding highlights an important nuance in the original paper's convergence guarantees, which hold in the limit but are sensitive to exploration strategy in risk-sensitive environments.

Future work could extend this study in several directions. The most natural extension, and the focus of Phase 2 of this project, is a direct comparison with SARSA and Expected SARSA. These on-policy algorithms are theoretically predicted to converge to the same safe CliffWalking route as observed here, but for fundamentally different reasons, making the behavioral comparison analytically meaningful. Additionally, applying Q-learning with linear function approximation to continuous state spaces would test the algorithm's scalability beyond the tabular setting.

11. Reproducibility Verdict

Classification: Partially Reproducible

The core algorithmic behavior described in Watkins & Dayan (1992) -convergence of the Q-function to a stable estimate, improvement of policy quality over training, and Bellman consistency of learned Q-tables is fully reproducible. All four environments produced results consistent with the paper's theoretical predictions.

The classification is 'partially' rather than 'fully' reproducible for two reasons. First, the paper does not provide sufficient hyperparameter detail to allow exact numerical replication without tuning. Second, the original experimental environments are no longer available, making direct numerical comparison of reward values impossible. The qualitative and theoretical claims of the paper are, however, strongly supported by the reproduction results.